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SPECIAL ARTICLE

Society of Critical Care Medicine Guidelines on Caring for Older Adults in the ICU

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RATIONALE: Older adults (those 65 years old or greater) compose a substantial proportion of the ICU population. As older adults with critical illness possess unique factors and considerations relevant to their care and outcomes, there is a need for evidence-based recommendations to guide critical care clinicians in the care of older ICU patients.

OBJECTIVE: The objective of this guideline is to develop evidence-based recommendations addressing the care of older adults during and after critical illness.

DESIGN: The American College of Critical Care Medicine Board convened a 22-member interprofessional panel, comprising physicians, advanced practice providers, nurses, a pharmacist, physical therapist, occupational therapist, and a patient representative. The panel included two expert methodologists specialized in developing evidence-based recommendations in alignment with the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) methodology. Conflict-of-interest policies were strictly followed during all phases of guideline development including task force selection and voting.

METHODS: The panel members prioritized five Population, Intervention, Comparator, and Outcomes questions. A systematic review was conducted for each question to identify the best available evidence, synthesize the evidence and assess the certainty of evidence using GRADE. The evidence-to-decision framework was used to formulate recommendations.

RESULTS: The panel generated two conditional recommendations and three "no recommendation" statements. The conditional recommendations are: 1) We suggest a geriatric model of care for all older adults admitted to the ICU and 2) We suggest not using antipsychotic medications for the prevention of delirium in older adults with critical illness. The three "no recommendation" statements are: 1) We make no recommendation regarding specialized post-ICU follow-up for older survivors of critical illness, 2) For older adults (age 65 and over) admitted to the ICU with vasodilatory shock, we make no recommendation with regard to targeting a mean arterial pressure (MAP) of 60–65 mm Hg as compared with usual care (MAP target > 65 mm Hg), and 3) We make no recommendation regarding the use of antipsychotic medication in the treatment of delirium in older adults with critical illness.

CONCLUSIONS: The guideline panel developed recommendations on caring for older adults during and after critical illness. Areas for future research were also identified during the guideline process.

KEYWORDS: clinical practice guideline; critical care; critical illness; geriatrics; intensive care unit; older adults

Older adults, defined as those 65 years old and older, have incurred more than half of ICU days for the past 25 years in the United States (1). With an aging population, including an expected doubling of this

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demographic worldwide by 2050, the proportion of older ICU patients is expected to increase (2). In light of these demographic shifts and the limited geriatrician workforce, it is imperative that all critical care clinicians are adept in the care of older ICU patients (3).

Care of the older ICU patient requires knowledge of factors and considerations unique to older adults. Conditions such as frailty or disability in functional activities, which are more commonly found in older than younger adults, are often stronger determinants of health outcomes than chronological age (4). Compared with younger patients, older patients are at increased risk of harm from common practices in the ICU and hospital, such as polypharmacy and immobility. They are also more vulnerable to delirium, particularly with any degree of pre-ICU cognitive impairment, highlighting the importance of routine delirium monitoring and avoidance of oversedation. In studies of health outcome prioritization, older adults rank "maintaining functional independence" as their most important health priority (over survival), necessitating the consideration of long-term functional recovery during ICU and post-ICU care (5).

Because they represent a large and growing demographic in the ICU, the unique considerations related to their care, and the lack of recommendations regarding the care of this complex and vulnerable population, we convened an expert panel to develop a clinical practice guideline on caring for older adults in the ICU.

METHODOLOGY

The American College of Critical Care Medicine, the consultative body of the Society of Critical Care Medicine (SCCM), commissioned a panel of experts to develop focused, evidence-based recommendations on caring for older adults in the ICU. This taskforce included physicians (including a geriatrician), advanced practice providers, pharmacists, nurses, physical therapists, occupational therapists, a patient representative, and other allied health and administrative professionals. Two methodologists from the Guidelines in Intensive Care Development and Evaluation group at McMaster University supported guideline development. To promote transparency and minimize risk of bias, members of the panel were required to disclose conflicts of interest (COI) in accordance with SCCM COI policy. Disclosures were updated at each phase of the guideline process and before every panel meeting. No author was restricted in their participation in developing, discussing, or voting for the guideline recommendations.

We considered a range of potential Population, Intervention, Comparator, and Outcomes (PICO) questions. **Table 1** presents the final PICO questions prioritized for guideline development; the **Supplemental Materials** (https://links.lww.com/CCM/H917) includes additional PICO questions that did not meet prioritization. The panel rated the relative importance of outcomes to identify those critical or important for decision-making (6). We conducted comprehensive and structured systematic reviews for each of the PICO questions with the support of a medical librarian. Literature searches were conducted in MEDLINE, Embase, and Cochrane CENTRAL Register of Controlled Trials (6). Preferred Reporting Items for Systematic Reviews and Meta-Analyses diagrams are provided in the Supplemental Materials (https://links.lww.com/CCM/H917) along with search strategies and queries. We used Covidence software (Veritas Health Innovation, Melbourne, Australia) (7) for dual and independent title and abstract screening, full-text review, and data extraction. Conflicts were resolved by discussion. We judged risk of bias for randomized controlled trials (RCTs) using the Cochrane Risk of Bias 2 tool (8). We performed meta-analyses where possible using Review Manager (RevMan) 5.3 software (The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark) (9) and summarized evidence narratively if not possible.

We used the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) process to develop recommendations (10) including the evidence-to-decision (EtD) framework using GradePro software (McMaster University and Evidence Prime, Canada) (11). For each PICO question, we considered the balance of desirable and undesirable effects, certainty of the evidence, patient values and preferences, resource considerations, feasibility, acceptability, and equity considerations (see Supplemental Materials, https://links.lww.com/CCM/H917 for EtD worksheets). Recommendations were approved if they received at least 80% of the vote by 75% or more of the panel.

We classified recommendations as either conditional or strong. A "Strong" recommendation is one

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TABLE 1.
Caring for Older Adults in the ICU Population, Intervention, Comparator, and Outcomes Questions

Question 1. Should Older Adults With Critical Illness Receive a Geriatrics Consultation or a Geriatric Model of Care Upon ICU Admission?

Population	Intervention	Comparator	Outcomes
Older adult patients admitted to any ICU	Geriatric consult or geriatric-specific model of care or care pathway	No geriatrics consult or no geriatric-specific model of care	Mortality
Subgroups to consider:			
Type of ICU (surgical vs. medical vs. CV surgery)			Quality of life
Older (age ≥ 65) vs. oldest old (age ≥ 80)			Disability in functional activities post-hospital discharge
			Long-term cognitive impairment
			Duration of IMV
			Ability to perform ADLs at long-term follow-up
			Discharge to nursing home (not previously at nursing home)
			Delirium
			Hospital LOS
			Depression, PTSD, anxiety symptoms

Question 2. Should Older Patients Who Survive Critical Illness Be Referred to Specialized Post-ICU Outpatient Follow-Up?

Population	Intervention	Comparator	Outcomes
Older adult patients who survive their ICU stay	Referral to specialized post-ICU follow-up	No specialized post-ICU follow-up	Mortality
Subgroups to consider:			Quality of life
Type of ICU (surgical vs. medical vs. CV surgery)			Disability in functional activities post-hospital discharge
Older (age ≥ 65) vs. oldest old (age ≥ 80)			Long-term cognitive impairment
Rehabilitation vs. just clinic			Duration of IMV
			Ability to perform ADLs at long-term follow-up
			Discharge to nursing home (not previously at nursing home)
			Delirium
			Hospital LOS
			Depression, PTSD, anxiety symptoms

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TABLE 1. (*Continued*)

Caring for Older Adults in the ICU Population, Intervention, Comparator, and Outcomes Questions

Question 3. Should We Aim for a Lower MAP Target (60–65 mm Hg) As Compared With Usual Care (Typically > 65 mm Hg) in Older Patients Admitted to the ICU With Vasodilatory Shock?

Population	Intervention	Comparator	Outcomes
Older adult patients admitted to ICU with shock	Lower MAP target (e.g., 60–65 mm Hg)	Usual care (typically MAP > 65 mm Hg)	Mortality
Subgroups to consider:			
Type of vasodilatory shock (septic vs. other)			Quality of life
Specific MAP target			Disability in functional activities post-hospital discharge
Older (age ≥ 65) vs. oldest old (age ≥ 80)			Long-term cognitive impairment
			Duration of IMV
			Ability to perform ADLs at long-term follow-up
			Discharge to nursing home (not previously at nursing home)
			Delirium
			Hospital LOS
			Depression, PTSD, anxiety symptoms

Question 4. Should Older Patients Admitted to the ICU Receive Antipsychotics for Prevention of Delirium?

Population	Intervention	Comparator	Outcomes
Older adult patients admitted to any ICU	Antipsychotics for prevention of delirium	No antipsychotic for prevention of delirium	Mortality
Subgroups to consider:			Quality of life
Type of ICU (surgical vs. medical vs. CV surgery)			Disability in functional activities post-hospital discharge
Older (age ≥ 65) vs. oldest old (age ≥ 80)			Long-term cognitive impairment
Type of antipsychotic			Duration of IMV
Those at highest risk for delirium vs. all older adults			Ability to perform ADLs at long-term follow-up
			Discharge to nursing home (not previously at nursing home)
			Delirium
			Hospital LOS
			Depression, PTSD, anxiety symptoms

(*Continued*)

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TABLE 1. (*Continued*)

Caring for Older Adults in the ICU Population, Intervention, Comparator, and Outcomes Questions

Question 5. Should Older Patients Admitted to the ICU Receive Antipsychotics for Treatment of Delirium?			
Population	**Intervention**	**Comparator**	**Outcomes**
Older adult patients admitted to any ICU with delirium	Antipsychotic for treatment of delirium	No antipsychotic for treatment of delirium	Mortality
Subgroups to consider:			Quality of life
Type of ICU (surgical vs. medical vs. CV surgery)			Disability in functional activities post-hospital discharge
Older (age ≥ 65) vs. oldest old (age ≥ 80)			Long-term cognitive impairment
Type of antipsychotic			Duration of IMV
Hypoactive vs. hyperactive delirium			Ability to perform ADLs at long-term follow-up
			Discharge to nursing home (not previously at nursing home)
			Delirium
			Hospital LOS
			Depression, PTSD, anxiety symptoms

ADLs = activities of daily living, CV = cardiovascular, IMV = invasive mechanical ventilation, LOS = length of stay, MAP = mean arterial pressure, PTSD = post-traumatic stress disorder.

that clinicians should follow for nearly all patients. A "Conditional" recommendation reflects a closer balance between benefits and harms and/or a lower certainty in evidence. Corresponding recommendations are worded as "We recommend" and "We suggest," respectively. **Table 2** describes the implications for recommendations.

RECOMMENDATIONS

We suggest a geriatric model of care for all older adults admitted to the ICU (conditional recommendation, very low certainty).

To address this question, we incorporated data examining a geriatric-specific model of care or formal geriatric consultation in patients with critical illness. We found two RCTs and one pilot trial (12–14).

One RCT investigated a clinical decision support system incorporating early consultation of a geriatrician and other aspects of age-friendly care, such as discontinuing restraints and urinary catheters, and stopping unnecessary anticholinergic drugs to reduce the incidence of delirium (12). There was an uncertain effect of this clinical decision support system incorporating geriatric models of care on mortality (relative risk [RR], 0.40; 95% CI, 0.08–1.90; very low certainty). The second RCT, investigating systematic transfer of older adults to a geriatric ward after ICU discharge (13) showed an uncertain effect on disability in functional activities after hospital discharge (mean difference on the Barthel index, 11.0 points higher in the group transferred to the geriatric ward, indicating less disability in activities of daily living [ADLs]; 95% CI, 6.7 lower to 28.7 higher; very low certainty). This trial also found an uncertain effect on the ability to perform ADLs 6 months after hospital discharge (RR, 0.94; 95% CI, 0.56–1.56; very low certainty). Pooled analysis from these RCTs found geriatric models of care had an uncertain effect on discharge to a nursing home (RR, 0.88; 95% CI, 0.64–1.21; very low certainty) and incidence of in-hospital delirium (RR, 0.59; 95% CI, 0.24–1.46; very low certainty). The impact of incorporating geriatric models of care on hospital length of stay was also uncertain (mean difference [MD], 2.3 d longer; 95% CI, 3.79 d shorter to 8.39 d longer; very low certainty) (12). A pre-/post-intervention pilot trial of a

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TABLE 2.
Implications of Strong and Conditional Recommendations

Target Audience	Strong Recommendation	Conditional Recommendation
For patients	Most individuals in this situation would want the recommended course of action and only a small proportion would not	The majority of individuals in this situation would want the suggested course of action, but many would not
For clinicians	Most individuals should receive the recommended course of action. Adherence to this recommendation according to the guideline could be used as a quality criterion or performance indicator. Formal decision aids are not likely to be needed to help individuals make decisions consistent with their values and preferences	Recognize that different choices will be appropriate for different patients and that you must help each patient arrive at a management decision consistent with her or his values and preferences. Decision aids may well be useful helping individuals making decisions consistent with their values and preferences. Clinicians should expect to spend more time with patients when working toward a decision
For policy makers	The recommendation can be adapted as policy in most situations including for the use as performance indicators	Policymaking will require substantial debates and involvement of many stakeholders. Policies are also more likely to vary between regions. Performance indicators would have to focus on the fact that adequate deliberation about the management options has taken place

Source: Grading of Recommendations, Assessment, Development, and Evaluation Handbook (10).

three-component ICU intervention based on geriatric models of care (treatment of hearing impairment with a portable amplifying device, deprescribing potentially inappropriate medications, and occupational therapy delivered as a default, all superimposed on usual care including the ABCDEF bundle) suggested that the intervention reduced development of delirium (14). A summary of findings from these studies is presented in the Supplemental Materials (https://links.lww.com/CCM/H917).

The overall certainty of evidence for this recommendation was very low due to issues with indirectness, imprecision, and risk of bias. Despite these uncertain effects, the panel judged the desirable effects of geriatric consultation or models of care as small-to-moderate based on indirect evidence from trauma, surgical, burns, and other in patient locations (15, 16). Furthermore, many of the point estimates fall on the side of benefit with inclusion of geriatric principles. Inclusion of geriatric principles and incorporation of geriatric models of care aligns with the Age-Friendly Health Systems initiative to incorporate the 4Ms (What Matters Most, Medications, Mobility, and Mentation) into the care of older inpatients (17). The panel also raised that some of the outcomes examined (e.g., mortality, length of stay) may be less important or at least equally important to older adults than living independently and cognitively intact (5, 18). Finally, the panel judged the undesirable effects of geriatric consultation or models of care as trivial, noting that they could not foresee how these would harm patients.

The cost-effectiveness of implementing geriatric models of care in older adults with critical illness was uncertain. While the addition of geriatric practitioners to the care team could increase costs, it is unknown whether implementation of geriatric models of care in the ICU would affect costs. Furthermore, decreased length of stay and delirium could improve patient outcomes and therefore reduce costs related to acute and post-acute healthcare. The panel had concerns related to health equity given the heterogeneity in availability of geriatricians across ICUs; however, their presence could also increase health equity for older adults. Finally, the panel judged that geriatric models of care are probably acceptable to key interest-holders and feasible to implement in clinical practice (see EtD in Supplemental Materials, https://links.lww.com/CCM/H917).

The panel was intentionally ambiguous regarding the specific geriatric models of care to be implemented as part of this recommendation and acknowledged interventions may vary depending on hospital, resources, and expertise available. At a minimum, the included studies intentionally incorporated geriatric

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principles into the care of older adults, such as removal of unnecessary tethering devices (urine catheters and restraints), addressing hearing impairment, and a focus on functional and cognitive outcomes through occupational therapy.

We make no recommendation regarding specialized post-ICU follow-up for older survivors of critical illness (conditional recommendation, low certainty).

Twenty-four RCTs were found, but none were conducted specifically in older adults. After contacting each study author, we obtained older adult subgroup data for five RCTs, which composed the data included in this evidence summary. The most common type of intervention was a comprehensive outpatient multidisciplinary clinic (19–21), but also included outpatient physical therapy (22) and an outpatient nurse-led program to improve psychological health after intensive care (23).

Pooled analysis found an uncertain effect of specialized post-ICU follow-up compared with no specialized follow-up on mortality (RR, 1.07; 95% CI, 0.62–1.84; very low certainty). Likewise, post-ICU follow-up may have no effect on health-related quality of life. Meta-analysis demonstrated that post-ICU follow-up may have no effect on the physical component of the 36-Item Short Form Health Survey (SF-36; MD, 1.01 points; 95% CI, 3.41 points less to 5.44 points more; low certainty) and uncertain effects on the mental component of the SF-36 (MD, 0.51 points less; 95% CI, 9.73 points less to 8.71 points more; very low certainty). Pooled analysis from studies using the EQ-5D may have no effect on health-related quality of life (MD, 0.04 points higher; 95% CI, 4.38 points lower to 4.45 points higher; low certainty). Last, current models of specialized post-ICU follow-up may have no effect on anxiety and depression symptoms as assessed using the Hospital Anxiety and Depression Scale (MD, 0.05 points higher; 95% CI, 2.23 points lower to 2.33 points higher; low certainty). A summary of findings is presented in the Supplemental Materials (https://links.lww.com/CCM/H917).

We judged the overall certainty of evidence as low due to inconsistency, indirectness, and imprecision. The panel agreed it was difficult to draw a meaningful conclusion with respect to the balance between desirable and undesirable effects of specialized post-ICU care for older adults based on the uncertain evidence and the heterogeneity in the multicomponent post-ICU follow-up interventions. Many of the included RCTs did not specify the exact components of the intervention or comment on intervention fidelity. There was some concern that the outcomes studied (e.g., mortality) may not be affected by the intervention while other outcomes that are potentially meaningful to patients and their caregivers, such as return-to-home, patient engagement, and return to social roles or activities, were not reported.

The panel also concluded that future health services research studies are needed to address important gaps related to the benefits of specialized post-ICU follow-up care. These include an evaluation of required resources and cost-effectiveness for the intervention, which were not addressed in the included studies. For example, clinic space and clinician time are needed to operate these clinics and could be resource intensive. Alternatively, if long-term follow-up could be a means by which to restore independent function for older adults or reduce hospital readmissions, there is potential for health system cost savings. The acceptability and feasibility of specialized post-ICU follow-up and effects on health equity are uncertain (see EtD in Supplemental Materials, https://links.lww.com/CCM/H917).

Given the large degree of uncertainty related to the above factors, the panel chose not to make a recommendation for or against long-term follow-up. Nevertheless, this uncertainty represents an area for future research, with a focus on older adults in general and by addressing the unique needs of this heterogeneous and vulnerable population. This includes the extent to which older adults with frailty, disability, and cognitive impairment may benefit from specialized post-ICU follow-up.

For older adults (age 65 and over) with vasodilatory shock, we make no recommendation with regard to targeting a mean arterial pressure (MAP) of 60–65 mm Hg as compared with usual care (MAP target > 65 mm Hg) (conditional recommendation, very low certainty).

We included one RCT comparing a lower MAP target (60–65 mm Hg) to usual care (at the discretion of the treating clinician, typically > 65 mm Hg) that was conducted in older adults with critical illness with vasodilatory shock (24). In keeping with the trial

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objective, patients in the lower MAP target group had less exposure to vasopressors. The trial demonstrated an uncertain effect of MAP targets on 28-day mortality (RR, 0.94; 95% CI, 0.84–1.04; very low certainty) and mortality at longest follow-up (RR, 0.93; 95% CI, 0.85–1.03; very low certainty) compared with higher MAP targets. In a prespecified secondary analysis, after adjustment for baseline variables, the odds ratio for 90-day mortality was 0.82 (95% CI, 0.68–0.98). There was an uncertain effect on quality of life as assessed using the EQ-5D-5L (MD, 0.01 points lower; 95% CI, 0.04 points lower to 0.03 points higher; moderate certainty) or cognitive impairment assessed with the Informant Questionnaire on Cognitive Decline in the Elderly score (MD, 0.13 points higher; 95% CI, 0.03 points lower to 0.29 points higher; moderate certainty). Lower MAP targets had an uncertain effect on the duration of mechanical ventilation (MD, 0.03 d less; 95% CI, 1.03 d less to 0.43 d more; moderate certainty), hospital length of stay (MD, 0.00 d; 95% CI, 1.91 d shorter to 1.91 d longer; low certainty), risk of serious adverse events (RR, 1.07; 95% CI, 0.79–1.45; low certainty), or need for renal replacement therapy (RR, 1.00; 95% CI, 0.87–1.15; low certainty). A summary of findings is presented in the Supplemental Materials (https://links.lww.com/CCM/H917).

The panel judged the desirable effects of lower MAP targets as uncertain, and the certainty of evidence as very low due to imprecision. Although undesirable effects are hard to capture, with the lack of long-term follow-up, the panel felt the undesirable effects were likely trivial. Cost-effectiveness data indicated targeting lower MAP targets was cheaper at 90 days (probability of cost-effectiveness for lower MAP targets = 70%) but the effect was attenuated at 1 year (40% probability of cost-effectiveness) (25). The panel concluded that using fewer vasopressors would correlate with beneficial effects such as fewer central lines, arterial lines, and shorter durations of ICU and hospital stay. The impact on health equity remains relatively unknown, but the panel recognized that lower MAP targets may offer practical advantages in low resource settings where vasopressors are scarce or unavailable. Lower MAP targets were judged by the panel as probably acceptable to key interest-holders, although some clinicians may be uncomfortable with the lower MAP if inconsistent with their current practice. Last, the panel determined that lower MAP targets would be feasible to implement in clinical practice if supported by the evidence (see EtD in Supplemental Materials, https://links.lww.com/CCM/H917). Due to the very low certainty of evidence, with only one trial published on this topic, the panel could not make a recommendation for the intervention and agreed that further research was needed.

We suggest not using antipsychotic medications for the prevention of delirium in older adults with critical illness (conditional recommendation, very low certainty).

Two antipsychotic drugs, haloperidol and quetiapine, were investigated in RCTs for the prevention of delirium in adults with critical illness (26–28). No study reported data on older adults nor were subgroup data for this demographic available. Three RCTs investigating the use of haloperidol or quetiapine for the prevention of delirium in adults with critical illness (not exclusively those over 65) were captured and analyzed for this question, with a downgrading of the certainty rating in the domain of imprecision given this data was not specific to the target population. The mean age of participants in these three studies ranged from 57 to 66.4 years, with one study excluding those over age 85 years.

Pooled analysis found that prophylactic antipsychotic administration had an uncertain effect on the incidence (RR, 0.92; 95% CI, 0.60–1.41) and duration of delirium (1.07 d less; 95% CI, 1.56 d less to 0.59 d less). One RCT (26) found no difference in delirium- and coma-free days (MD, 0 d; 95% CI, 0.77 d less to 0.77 d more; moderate certainty) between prophylactic antipsychotics and placebo. Effects on mortality (RR, 0.99; 95% CI, 0.80–1.23), duration of invasive mechanical ventilation (MD, 2.31 less days; 95% CI, 6.12 d less to 1.49 d more), ICU length of stay (MD, 0.70 less days; 95% CI, 3.02 d less to 1.54 d more), hospital length of stay (MD, 1.96 less days; 95% CI, 6.90 d less to 2.98 d more), and adverse effects (RR, 1.37; 95% CI, 0.31–6.00) were likewise uncertain. A summary of findings is presented in the Supplemental Materials (https://links.lww.com/CCM/H917).

The overall certainty of evidence for most of the outcomes was very low due to indirectness, risk of bias, inconsistency, and imprecision. While the included studies enrolled older adults, these trials did not analyze this population separately, which limited our ability to draw direct population-specific conclusions.

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Imprecision and clinical heterogeneity led to challenges assessing the balance between desirable and undesirable effects and ultimately led to a conditional recommendation. The only outcome that had moderate certainty, delirium- and coma-free days, did not show any difference with antipsychotics. While it could be the case that antipsychotics truly had no effect on this outcome, panelists also expressed concern that these drugs could reduce hyperactive symptoms, which aid in delirium detection. Without more nuanced data, this could not be explored in further detail. While the included studies did not demonstrate important adverse effects, the panel acknowledged widespread issues with inappropriate antipsychotic prescription in ICU patients and that these drugs are often continued after ICU discharge or hospital discharge (29). Because antipsychotics are included in high-risk medication lists for older adults and have been clearly associated with adverse events in other studies of older survivors of critical illness, the panel was concerned that use of these medications could lead to potential harm in older adults during or after critical illness (29–31).

The required resources and cost-effectiveness of prophylactic antipsychotics are uncertain; however, these drugs are generally inexpensive, widely available and feasible to administer. The panel judged that the impact on health equity was uncertain and acknowledged that the detection and management of delirium differs based on social determinants of health, race, ethnicity, and language (32–34). Given the known risks of antipsychotics in older adults in general, as well as negative perceptions of sedative effects for patients and families, the routine use of prophylactic antipsychotics for older adults with critical illness in the absence of delirium may be unacceptable to stakeholders (see EtD in Supplemental Materials, https://links.lww.com/CCM/H917).

We make no recommendation regarding the use of antipsychotic medication in the treatment of delirium in older adults with critical illness (conditional recommendation, low certainty).

Of six RCTs investigating antipsychotic medications for the treatment of delirium in adults with critical illness, we obtained subgroup data on adults 65 years old and older from two RCTs that investigated haloperidol compared with placebo (35, 36).

Pooled analysis found that haloperidol may reduce mortality (RR, 0.84; 95% CI, 0.68–1.03; low certainty) in older adults with critical illness experiencing delirium. However, effects were uncertain on delirium- and coma-free days (MD, 2.48 d more; 95% CI, 2.59 d less to 7.55 d more; very low certainty) and hospital length of stay (MD, 1.07 d more; 95% CI, 2.30 d less to 4.44 d more; very low certainty). One RCT showed that haloperidol may have no effect on duration of mechanical ventilation (MD, 0 d; 95% CI, 1.32 d less to 1.32 d more; low certainty) (36). The other RCT demonstrated haloperidol had an uncertain effect on days alive and free of mechanical ventilation (MD, 9.50 d more; 95% CI, 0.47 d more to 18.53 d more; low certainty) and days alive without coma or delirium (MD, 7.00 d more; 95% CI, 2.08 d less to 16.08 d more) (35). Haloperidol was probably not associated with an increased risk of adverse events (RR, 0.78; 95% CI, 0.49–1.26; moderate certainty). A summary of findings is presented in the Supplemental Materials (https://links.lww.com/CCM/H917).

The possible reduction in mortality was carefully considered by the panel. While the included studies only evaluated haloperidol, one of these trials bundled the antipsychotic treatment with other interventions, which contributed to indirectness and may have at least partially explained the benefit seen (36). Also, the fact that this beneficial effect did not translate into other patient-important outcomes tempered the panel's enthusiasm. As with the prophylaxis recommendation, adverse events associated with antipsychotics are probably under-reported across studies given what is known from other studies including older adults.

Overall, the panel judged the desirable effects of antipsychotic treatment in older adults with critical illness as unknown while the undesirable effects were judged as trivial with an unclear balance. The required resources and cost-effectiveness of treating older adults with critical illness who experience delirium with antipsychotics are unknown, with similar considerations to the prophylaxis recommendation that the drugs are widely available, inexpensive, and feasible to implement. The panel judged that the impact on health equity was unknown and acknowledged that delirium may be managed differently based on social determinants of health, race, ethnicity, and language (see EtD in Supplemental Materials, https://links.lww.com/CCM/H917).

In light of important ongoing uncertainty for multiple crucial aspects of the EtD framework (including

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TABLE 3.
Caring for Older Adults in the ICU Research Priorities

Topic	Research Priorities
Geriatric consultation or geriatric-specific model of care in the ICU	Studies that further investigate the impact of geriatric consultation or geriatric models of care in the ICU, with comparison to the standard of care
	Geriatric consultation and geriatric models of care may both fall under the framework of "Age-Friendly" care models, which are increasingly being implemented in health systems, particularly in the U.S. future studies of age-friendly care should evaluate the impact of clearly specified components (i.e., the 4Ms of Age-Friendly Care: What Matters, Medication, Mentation, Mobility) compared with standard of care. Reporting fidelity of delivery of these components should be central to interpreting intention-to-treat and per-protocol analyses of efficacy
	Outcomes in future studies should focus on the goals of geriatric consultation or geriatric models of care, mortality between 30 and 90 d, and patient-centered outcomes that matter most to older adults, such as short-and long-term functional and cognitive outcomes, maintaining community independence, and mobility
Specialized post-ICU outpatient follow-up	A core outcome set for older adults who survive critical illness is needed to ensure outcomes that are the most important to patients are gathered in future research
	Address heterogeneity of older adults by evaluating whether specialized follow-up (and related interventions) is of greatest benefit in those with vulnerability factors such as biological or social frailty
	Studies investigating acceptability of specialized geriatric care from multiple viewpoints (patient, family/caregiver, primary care, etc)
	Develop a conceptual model of ideal recovery for an older adult recovering from a critical illness to tailor interventions toward recovery
MAP targets in older patients admitted to ICU with vasodilatory shock	More high-quality trials investigating MAP targets in older adults with critical illness from vasodilatory shock to increase certainty of evidence
	Determine the effects of MAP targets on longer-term, patient-centered outcomes such as functional and cognitive outcomes
	Evaluate whether midodrine changes the balance of benefits and harms for MAP targets in future trials conducted in this population
Antipsychotics for prevention of delirium	Research exploring the balance of preventing delirium with the harm of exposing older adults to antipsychotic risks
	Studies identifying patients that may benefit most (i.e., subgroups) from prevention of delirium or patients at the highest risk for delirium are needed
Antipsychotics for treatment of delirium	Trials enrolling exclusively older adults with critical illness to investigate antipsychotic use for the treatment of delirium
	Research on long-term outcomes of antipsychotic use in older adults with critical illness
	Trials reporting the quality of implementation of nonpharmacological bundles when antipsychotics are used
	Investigations to improve the accuracy of delirium assessment in older adults with critical illness and thereby enhance detection of treatment effects
	Cost-effectiveness research and resource-use research
	Studies investigating acceptability of antipsychotic treatment from multiple viewpoints (patient, family/caregiver, primary care, etc)

MAP = mean arterial pressure.

undesirable effects), the panel chose to make no recommendation despite the possible positive effect on mortality. For this PICO question, and PICO question 2 (on follow-up clinics), we relied on available subgroup analysis from eligible RCTs. Not having subgroup data from all RCTs, not having stratified

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analyses, and not having individual patient data limited our ability to make more firm subgroup conclusions and was factored into GRADE certainty ratings. Future large-scale RCTs specific to older adults will be crucial to further evaluate the role of antipsychotics in this population with careful collection of patient-important outcomes, including long-term outcomes and adverse effects.

RESEARCH AGENDA

We developed a research agenda for the care of older adults in the ICU. During each EtD meeting, the panel identified research priorities for each PICO question and listed them in the EtD tables (Supplemental Materials, https://links.lww.com/CCM/H917). The research agenda is presented in **Table 3**.

CONCLUSIONS

We make several recommendations addressing care of older adults with critical illness and identify areas for future research in this population. Future iterations of this guideline will provide updates in these areas and address additional aspects of ICU care for older adults.

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